A volcano is, in fact, an aperture to an underground realm much vaster than the thin surface layer that humans inhabit, and far more hostile. The lava that erupts from a volcano is liquid rock—rock raised to its melting point, generally around 1000°C. The lava emerges from a hole in the Earth; as it cools and solidifies, it generates and later remakes the flanks of a volcanic mountain.

The most volcanically active locales on Earth tend to be along ridges on the ocean floor and island arcs—at the junction of two great plates of oceanic crust—either separating from each other, or one slipping under the other. On the seafloor there are long zones of volcanic eruptions—accompanied by swarms of earthquakes and plumes of abyssal smoke and hot water—that we are just beginning to observe with robot and manned submersible vehicles.

Eruptions of lava must mean that the Earth's interior is extremely hot. Indeed, seismic evidence shows that, only a few hundred kilometers beneath the surface, nearly the entire body of the Earth is at least slightly molten. The interior of the Earth is hot, in part, because radioactive elements there, such as uranium, give off heat as they decay; and in part because the Earth retains some of the original heat released in its formation, when many small worlds fell together by their mutual gravity to make the Earth, and when iron drifted down to form our planet's core.

The molten rock, or magma, rises through fissures in the surrounding heavier solid rocks. We can imagine vast subterranean caverns filled with glowing, red, bubbling, viscous liquids that shoot up toward the surface if a suitable channel is by chance provided. The magma, called lava as it pours out of the summit caldera, does indeed arise from the underworld. The souls of the damned have so far eluded detection.

Once the volcano is fully built from successive outpourings, and the lava is no longer spewing up into the caldera, then it becomes just like any other mountain—slowly eroding because of rainfall and windblown debris and, eventually, the movement of continental plates across the Earth's surface. "How many years can a mountain exist before it is washed to the sea?" asked Bob Dylan in the ballad "Blowing in the Wind." The answer depends on which planet we're